

PREDIX HER2 — patients with complete data across all five modalities

n = 110 (46 pCR)

DHP n = 59 (24 pCR)

T-DM1 n = 51 (22 pCR)

each modality-specific model is trained on all patients in whom that modality was measured, minus the patients of the current test fold, and is evaluated only on the complete-case cohort

Candidate features — a-priori biological curation, no outcome used

Clinical
5 metrics

Transcriptomic
42 metrics

Genomic
41 metrics

Proteomic
19 metrics

WSI
3 metrics

110 pre-defined metrics → fixed outcome-blind deduplication of co-amplified and near-identical features, applied once before any fold (−18) → 92 candidates

OUTER LOOP — stratified 5-fold cross-validation, RepeatedStratifiedKfold, seed 42

training

training

training

held-out
test fold

training

each fold is held out in turn (~80 % train / ~20 % test), stratified on pCR

the whole partition is redrawn × 200 repeats = 1,000 outer evaluations (pooled cohort)

arm-specific models: 5-fold × 100 repeats = 500 evaluations per arm,
with 3-fold inner cross-validation. No shuffle-split resampling at any stage.

WITHIN ONE OUTER TRAINING FOLD — fitted on training patients only

Preprocessing, in this order

near-zero-variance filter
median imputation, then standardisation
univariate screen: Mann–Whitney AUROC,
BH $q \leq 0.25$, keep 5–40 candidates
(clinical, WSI: ≤ 6 candidates, unscreened)
no in-fold correlation filter: redundancy is removed once, above, before any fold is drawn

Inner 5-fold cross-validation

Stage A the five classifier families below are compared under fixed grids; importances become percentile ranks and define the feature signature, capped at ≥ 5 pCR events per selected variable
Stage B the winning family is tuned by grid search, restricted to the discovered signature

ElasticNet logistic

RandomForest

ExtraTrees

HistGradBoost

SVM Linear

probabilities recalibrated by Platt scaling fitted inside the cross-validation
→ a leakage-safe out-of-fold probability for each of the five modalities

Late-fusion stacking

elastic-net logistic regression ($L1 + L2$, $l1_ratio = 0.5$) over the five calibrated probability streams;
the L1 term sets non-contributing modalities to exactly zero, giving interpretable modality weights

Applied unchanged to the held-out outer test fold

in every repeat each patient receives exactly one out-of-fold predicted probability;
no patient is ever scored by a model that saw them in training

Locked consensus model — aggregated over the 1,000 outer folds

for each modality the locked classifier is the family that won the most outer folds;
the signature is aggregated only over the folds that family won — one classifier, one matching signature
K = the median per-fold signature size; correlated DNA and RNA candidates are pooled to one representative

Reported performance

the metric (AUROC, AUPRC, Brier) is computed on each repeat's complete out-of-fold vector and averaged over the 200 repeats (100 per arm); 95 % confidence intervals are patient-level cluster bootstraps — 2,000 stratified resamples of patients, a resampled patient carrying all of its repeat predictions. Predictions are never averaged across repeats or across models.